Introduction paragraph

That this rule is a good classifier can be checked directly, and the check tells us what has to be studied. We used the open-source implementation of CLIP [1] provided by OpenCLIP [4], with the ViT-B/32 image encoder (a Vision Transformer [5] with 32×32 patches) and the checkpoint `laion2b_s34b_b79k` trained on the LAION-2B dataset [6, 7], to embed the 10,000 test images of CIFAR-100 [8] into $\mathbb{R}^{512}$ and, with the prompt "a photo of a {}", the names of its 100 classes; the code and data are available at `https://github.com/Mirkes/OneShotSeparabilityTheorem`. For every class z, the halfspace $\{x : (g(z), x) \geqslant \theta\}$ separates the images of class z from the images of all other classes with ROC AUC 0.98 on average and 0.91 at worst over the 100 classes (Section 8.3). Thus the text embedding $g(z)$ is a good normal vector for a halfspace that captures the class, and it is constructed from one object only—one description of the class—without any image of it. This is the situation studied in this paper: a hyperplane built from a single representative of a class, which must leave the representatives of all other classes on the other side. CLIP learns an embedding in which, for the captions it has seen, images rank correctly against sampled alternatives, and then one hopes that the halfspaces of m new labels will not capture each other's representatives. That hope is a geometric property of the learned embedding rather than a term of the loss, and it is the one-shot separability of the set of representatives defined in Section . The capacity of zero-shot classification—how many classes can be told apart by such a rule—is therefore governed by the one-shot separability of the embeddings of class descriptions, and in Section 8.3 we compare its behaviour on these data with the theory below: the rate at which the probability that one embedding violates the halfspace of another decays with the dimension, and the number of simultaneously usable labels, which follows the resulting bound until it is limited by the data rather than by the dimension.

8.3 One-shot separability of CLIP embeddings

We return to the embeddings described in the Introduction and compare them with the theory, again for $\alpha = 1$. The image embeddings were centred and whitened by principal components, so that the empirical covariance is the identity as in the isotropic setting of Sections –, and the dependence on dimension was examined by retaining the n leading principal components, ordered by eigenvalue. The label embeddings were treated in the same way for the dependence on dimension; the counts in the full 512-dimensional space, where whitening would amplify the low-variance directions, use the label embeddings projected onto the unit sphere and centred, which is the geometry in which CLIP uses them. For a set of N points and a map $X \mapsto c(X)$ we estimated $c(\mathbb{P}_n, 1)$ by the fraction of ordered pairs (X_i, X_j), $j \neq i$, with $(X_j - X_i, c(X_i)) \geqslant 0$, over 2,500 random anchors X_i and all $N - 1$ other points. Two maps were used: the universal map $c_i = h_1(x_i)$ of () and the Fisher map $c_i = x_i$; on these data they give indistinguishable results.

The zero-shot rule as a halfspace classifier. We first quantify the statement made in the Introduction; the numbers are collected in Table 1. On the raw embeddings, without normalisation or centring, the score $(g(z), f(u))$ separates the 100 images of class z from the 9,900 others with ROC AUC 0.982 on average over the 100 classes; 96 classes exceed 0.95, and the lowest value, 0.914, is attained by "flatfish". With a threshold θ_z chosen per class the halfspace $\{x : (g(z), x) \geqslant \theta_z\}$ captures 95.7% of its own images and 6.4% of the others; the optimal thresholds vary by only 9% across classes, and a single threshold common to all classes loses two points of balanced accuracy. The argmax rule over the 100 classes gives 72.9% accuracy with raw inner products and 75.8% with cosine similarity. The normal vector $g(z)$ of each halfspace is a proxy for the centroid of the class: after removing the translation common to all classes (the modality gap of [2]), the nearest

ROC AUC over the 100 classes	min	10%	25%	median	75%	90%	max
	0.914	0.964	0.976	0.985	0.992	0.996	0.999

Halfspace $\{x : (g(z), x) \geqslant \theta\}$	balanced accuracy	own images captured	other images captured
threshold θ_z chosen per class	0.946	95.7%	6.4%
one threshold θ for all classes	0.927	92.7%	7.3%

Table 1: The zero-shot rule as a halfspace classifier on the raw CLIP embeddings of the CIFAR-100 test set (10,000 images, 100 classes). Top: distribution over classes of the ROC AUC of the score $(g(z), f(u))$ for separating the images of class z from all others; the lowest values are "flatfish" (0.914), "man" (0.934), "worm" (0.945), the highest "bicycle" (0.999), "pickup truck" (0.999), "sunflower" (0.999). Bottom: balanced accuracy, recall and false-inclusion rate of the halfspace with the threshold maximising balanced accuracy, chosen per class or common to all classes. The argmax rule over the 100 classes has accuracy 72.9% with raw inner products and 75.8% with cosine similarity.

class centroid of every $g(z)$ is the centroid of class z for all 100 classes in cosine similarity, although the remaining class-specific difference is about half the norm of $g(z)$. The rest of this subsection concerns the question this raises: how many such halfspaces, each built from one representative, can coexist, and how this depends on the dimension.

Images. Figure 1 shows $\hat{c}(\mathbb{P}_n, 1)$ as a function of the number n of leading principal components. For $n \leqslant 60$ it decays as 0.839^n; in the notation of Table this is $c_{\mathbb{P}} \approx 0.84$ and $b_{\mathbb{P}} \approx 0.09$, slower than the normal (0.707) and Laplace (0.787) laws and just below the universal constant $C_{hl}(1) < 0.85$ of Theorem . From $n \approx 120$ on, the estimate is at the resolution of the estimator ($4 \cdot 10^{-8}$, one violating pair), and in the full 512-dimensional space all 9,999 distinct images are one-shot separable; the two violating ordered pairs there are a duplicated image, i.e. the coincidence of two sample points that no hyperplane can separate (cf. the footnote to Proposition). The figure is cut at $n = 250$, where the condition number λ_1/λ_n of the retained components reaches 100; it reaches 10 already at $n = 35$, which is marked for orientation. Beyond these points whitening divides the data by ever smaller eigenvalues ($\lambda_1/\lambda_n \approx 750$ at $n = 448$, and the last component is degenerate, $\lambda_{512} \approx 7 \cdot 10^{-10}$), so that the whitened trailing coordinates are increasingly dominated by whatever noise the encoder leaves in its low-variance directions; we therefore read the exponent from the well-conditioned range $n \leqslant 60$ and regard the counts at larger n only as confirmation that nothing changes when further components are added. Which family of Sections – does the leading subspace correspond to? Its coordinates are close to log-concave individually (median excess kurtosis 0.6, log-density concave on 85% of its range), its radius $\|X\|^2/n$ concentrates about as for a Gaussian sample of the same size, and the excess of the rate over the Gaussian one is entirely due to dependence between the coordinates: permuting each coordinate independently, which replaces the distribution by the product of its marginals, brings the rate down to 0.70^n. This dependence is directional rather than radial—the leading components carry the cluster structure of the classes, the first of them being bimodal—and a radial model with uniformly distributed directions and the empirical law of the norms underestimates the error (rate 0.67^n). In the language of [3], the embeddings have "sets of small volume with large probability", which is precisely what the density ratio in Proposition controls: the support is bounded, the uniform distribution in a ball is an admissible reference family, and the majorisation constant required to reach the observed rate is of the order $(0.84/0.71)^n \approx 1.2^n$, within the range $b < 1/\sqrt{c_{\mathbb{P}}} = \sqrt{2}$ in which the proposition yields exponential one-shot separability. Theorem , by contrast, does not apply as stated: the joint distribution is a mixture over classes and is not log-concave.

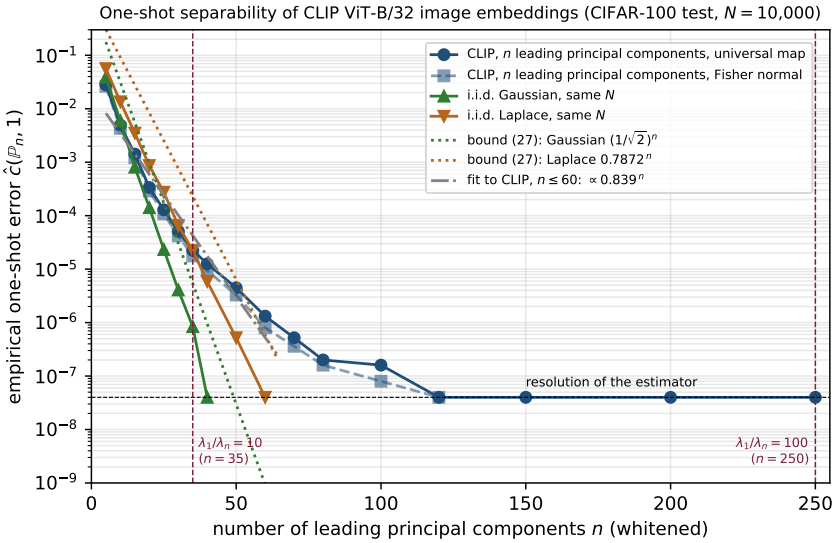


Figure 1: Empirical one-shot separability error $\hat{c}(\mathbb{P}_n, 1)$ of CLIP image embeddings (CIFAR-100 test set, $N = 10{,}000$) in the subspace of the n leading principal components after whitening, with i.i.d. Gaussian and Laplace samples of the same size, the bounds () for these two laws, and a fit $\propto 0.839^n$ to the CLIP data for $n \leqslant 60$. The vertical lines mark the dimensions at which the condition number λ_1/λ_n of the retained components reaches 10 and 100; the figure is cut at the latter.

Labels. For the text embeddings the decay with the number of leading components is at least as fast as for the images up to $n \approx 60$ and then levels off (Figure 2, left): for the full WordNet vocabulary at about $3 \cdot 10^{-7}$, for the ImageNet class names at three violating pairs out of 10^6, one of which is the class name "missile", which occurs twice in the ImageNet label list. What remains at the floor is semantic rather than geometric. In the full space, in the sphere-centred geometry, 1.98% of the WordNet nouns are not separable from some other noun (3,258 unordered pairs among 112,058 labels); keeping one lemma per WordNet synset reduces this to 1.11%, and excluding plant and animal taxa to 0.41%, with the universal map and the Fisher normal agreeing to within a few pairs. The remaining inseparable pairs are a term and its modification ("cassette"–"cassette tape", "icosahedron"–"regular icosahedron"), a hypernym inside the halfspace of its hyponym ("frog"–"leopard frog"), or ordinal variants ("phase I"–"phase II clinical trial"): the text encoder does not separate what the captions it was trained on did not separate. The capacity of the label space, i.e. the largest m for which m random labels of the cleaned vocabulary are all one-shot separable with probability 0.9, grows from 4 at $n = 10$ to 54 at $n = 30$, 425 at $n = 60$, 1,098 at $n = 100$ and 2,580 at $n = 250$ (Figure 2, right). Up to $n \approx 60$ it follows the bound $m < \sqrt{\delta/\hat{c}}$ of Proposition to within 10%, so that on these data the union bound in the proposition is not loose; beyond that the capacity exceeds the bound, because the remaining violations are concentrated on the few labels that have a near-duplicate, whereas the bound treats them as spread over all pairs. Two remarks connect this to the preceding sections. The floor is the coincidence of sample points excluded in Theorem (a), in a softened form; on the sphere itself, at $\alpha = 1$, even near-duplicate labels are separable by the tangent hyperplane, and the floor is a statement about margins, appearing for $\alpha < 1$. And the agreement of the capacity with $\sqrt{\delta/\hat{c}}$ in the well-conditioned range bears on the gap between m and m^2 discussed after Theorem : in the regime of these data the quadratic dependence

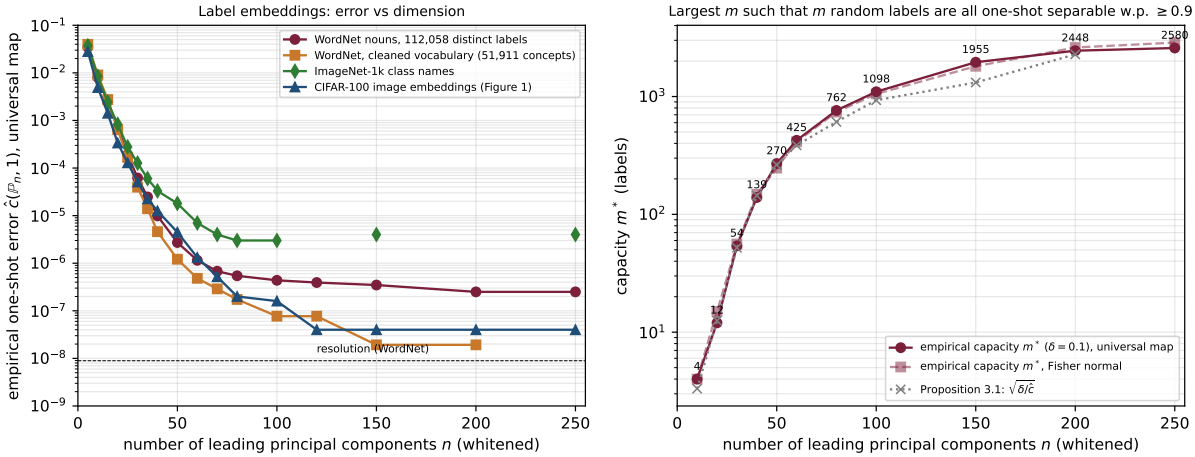


Figure 2: Left: $\hat{c}(\mathbb{P}_n, 1)$ for the embeddings of the 112,058 distinct WordNet nouns, of the cleaned vocabulary of 51,911 concepts and of the 1,000 ImageNet class names, in the subspace of the n leading whitened principal components, compared with the image embeddings of Figure 1; the label curves level off at the resolution of the estimator or at the level of semantic near-duplicates. Right: the largest m such that m random labels of the cleaned vocabulary are all one-shot separable with probability 0.9 (universal map and Fisher normal), against the bound $\sqrt{\delta/\hat{c}}$ of Proposition .

of Proposition is the correct one.

Proposition for Section 3

For rotation invariant distributions the minimiser in (8.3) is known explicitly, and $c(\mathbb{P}_n, \alpha)$ has been computed.

Proposition 8.1. *Let $\mathbb{P}_n$ be rotation invariant in the sense of (8.3), or more generally the law of RU with U uniformly distributed on the unit sphere and $R \geqslant 0$ independent of U. Then for every $\alpha \in (0, 1]$ and $x \neq 0$ the infimum in (8.3) is attained at the Fisher halfspace $\{z : (z - \alpha x, x) \geqslant 0\}$, so that $c(\mathbb{P}_n, \alpha)$ equals the probability that two i.i.d. points X, Y from $\mathbb{P}_n$ satisfy $\alpha(X, X) \leqslant (X, Y)$, i.e. form an α-Fisher inseparable pair. Consequently $\mathbb{P}_n$ admits exponential α-one-shot separability algorithmically with the Fisher map $\mathcal{A}(n, X) = \{z : (z - \alpha X, X) \geqslant 0\}$ and $\epsilon = 0$, and*

$$c(\mathbb{P}_n, \alpha) = \frac{\alpha}{B\left(\frac{n-1}{2}, \frac{1}{2}\right)} \int_0^{1/\alpha} \left(1 - \alpha^2 t^2\right)^{\frac{n-3}{2}} \mathbb{P}\left[\frac{\|X\|}{\|Y\|} \leqslant t\right] dt,$$

which is the quantity $p(n, \alpha)$ of [3, Proposition 6 and eq. (46)]; in particular, the optimal bound on m in [3, Theorem 14], derived there for α-Fisher separability, is by Theorem 8.3 necessary and sufficient for α-linear separability of exponentially many i.i.d. points as well.

Proof. Every closed halfspace whose boundary contains αx is $\boldsymbol{H}_u = \{z : (z, u) \geqslant \alpha(x, u)\}$ for some unit vector u. By rotation invariance, $\mathbb{P}_n(\{z : (z, u) \geqslant s\}) =: G_n(s)$ does not depend on u and is non-increasing in s, while $\alpha(x, u) \leqslant \alpha|x|$ with equality for $u = x/|x|$; hence $\mathbb{P}_n(\boldsymbol{H}_u) \geqslant G_n(\alpha|x|) = \mathbb{P}_n(\boldsymbol{H}_{x/|x|})$, and $\boldsymbol{H}_{x/|x|}$ is the Fisher halfspace. Taking the expectation over $X \sim \mathbb{P}_n$ in (8.3) gives $c(\mathbb{P}_n, \alpha) = \mathbb{P}[\alpha(X, X) \leqslant (X, Y)]$, since for fixed X the $\mathbb{P}_n$-measure of its Fisher halfspace is the conditional probability of this event. The integral formula is [3, Proposition 6], and the last

assertion follows from Theorem 8.3 because both separability notions are governed by the same quantity. □

Explicit expressions for $\mathbb{P}[\|X\|/\|Y\| \leqslant t]$, and hence for $c(\mathbb{P}_n, \alpha)$, are given in [3, Sections 5.1–5.4] for the uniform distributions in a ball and in a spherical layer, for the standard normal and for the multivariate exponential distributions; for a dataset, $\mathbb{P}[\|X\|/\|Y\| \leqslant t]$ is estimated by the empirical distribution of the ratio of two norms, which makes the formula a convenient reference family for Proposition .

References

[1] Radford et al. 2021.

[2] Liang et al. 2022.

[3] Grechuk et al. 2021.

[4] Ilharco et al. 2021.

[5] Dosovitskiy et al. 2021.

[6] Schuhmann et al. 2022.

[7] Cherti et al. 2023.

[8] Krizhevsky 2009.

[9] Miller 1995.